Math 421

Monday, December 1

Announcements

- Drop-In Hours
  - Wednesday 12-1 pm VV 311
  - Thursday 9-11 am VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
  - Course Assistant: Th 4-7 Table 4
- Homework 10 due Friday (last HW!)

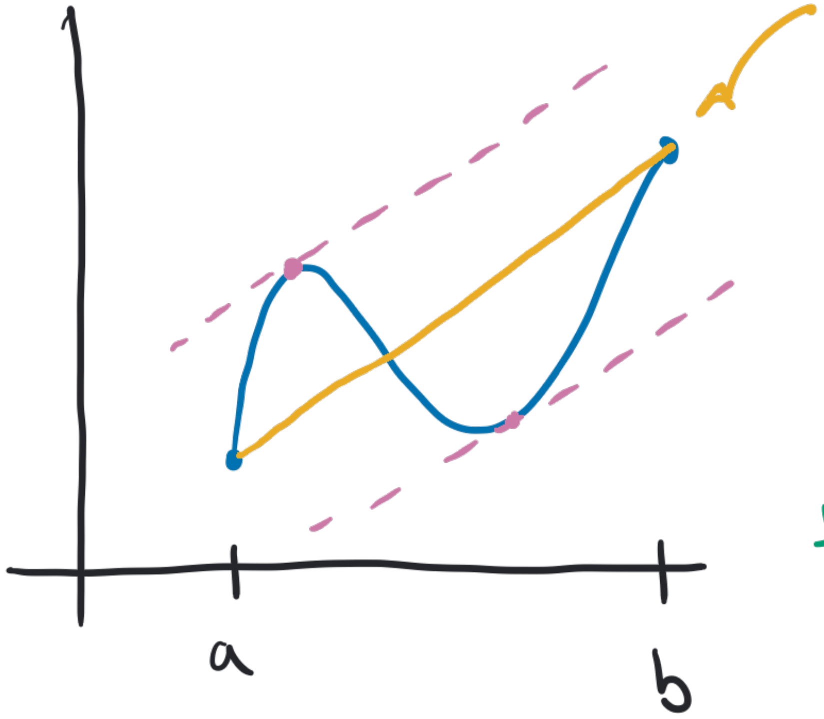
Rolle's Theorem. If g cont. on $[a,b]$, diff on (a,b) $g(a) = g(b)$, then there exists $x_0 \in (a,b)$ such that $g'(x_0) = 0$.

Mean Value Theorem

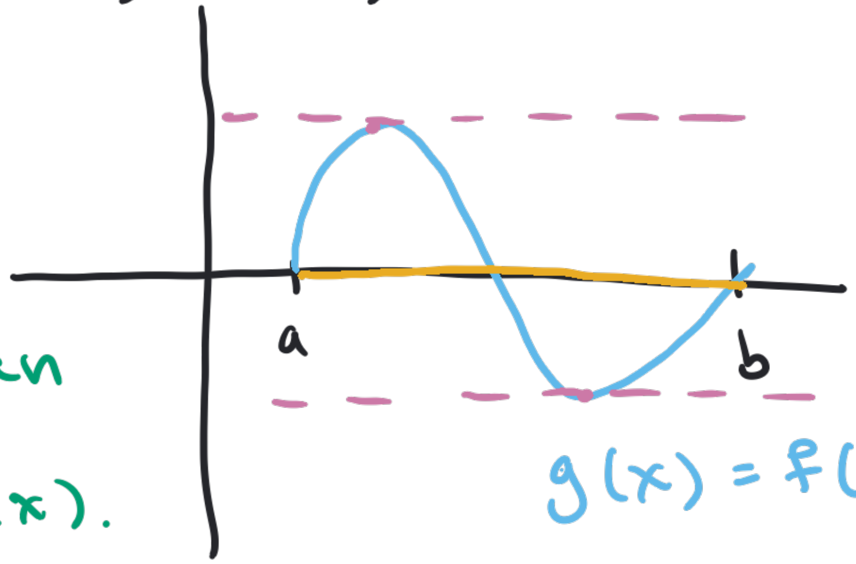
Theorem 29.3. If f is continuous on $[a, b]$ and differentiable on (a, b), then there is a number $x_0 \in (a, b)$ such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}$$

$$y = \frac{f(b) - f(a)}{b - a}(x - a) + f(a)$$



take the difference betwn the line & $f(x)$.



$g(x) = f(x) - \text{line}.$

Proof of MVT

Proof. Define $g(x) = f(x) - \left[\frac{f(b)-f(a)}{b-a}(x-a) + f(a)\right]$. Since f is continuous on $[a,b]$ and differentiable on (a,b), so is g.

Note that

$$g(a) = f(a) - \left[\frac{f(b)-f(a)}{b-a}(a-a) + f(a)\right] = 0$$ and

$$g(b) = f(\cancel{b}) - \left[\frac{\cancel{f(b)} - \cancel{f(a)}}{\cancel{b-a}}(\cancel{b-a}) + \cancel{f(a)}\right] = 0$$ so $g(a) = g(b)$.

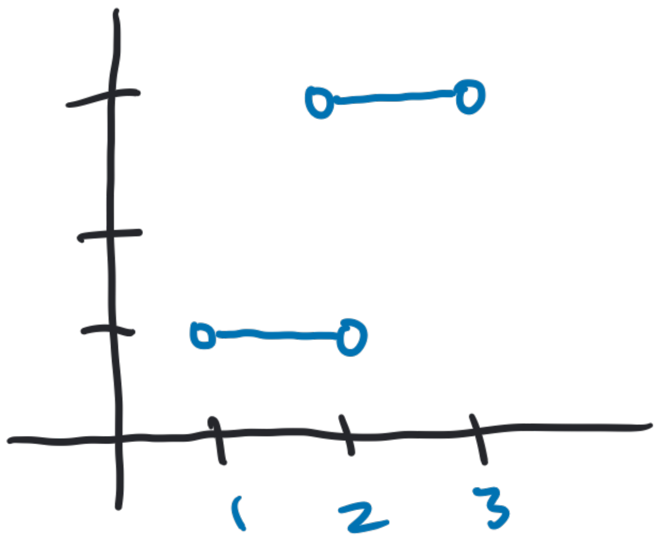
Therefore, by Rolle's Theorem, there exists $x_0 \in (a,b)$ such that $g'(x_0) = f'(x_0) - \frac{f(b)-f(a)}{b-a} = 0$ and so $f'(x_0) = \frac{f(b)-f(a)}{b-a}$.

Constant Functions

Corollary 29.4. If f is defined on (a,b) and $f'(x)=0$ for all $x \in (a,b)$, then f is constant on (a,b).

True/False: If $f'(x)=0$ for all $x \in \mathrm{dom}(f)$ then f is constant.

FALSE!



dom $f = (1,2) \cup (2,3)$

Proof of Corollary

Corollary 29.4. If f is defined on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b).

Proof. Assume f is defined on (a,b) and $f'(x)=0$ for all $x \in (a,b)$, and, by way of contradiction, that f is not constant on (a,b). Then there exist $x_1, x_2 \in (a,b)$ with $a < x_1 < x_2 < b$ and $f(x_1) \neq f(x_2)$. Since f is cont. on $[x_1, x_2]$ and diff. on (x_1, x_2). By the MVT, there exists $x_0 \in (x_1, x_2)$ such that $f'(x_0) = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1} \neq 0$. This contradicts that $f'(x) = 0$ for all $x \in (a,b)$. Therefore, f is constant.

Activity – Prove the Corollary

Corollary 29.5. If f and g are defined on (a, b) and $f'(x) = g'(x)$ for all $x \in (a, b)$, then there is some number c such that $f(x) = g(x) + c$ for all $x \in (a, b)$.

Proof. Define $h(x) = f(x) - g(x)$. Then h is defined on (a,b) and $h'(x) = 0 = f'(x) - g'(x)$ for all $x \in (a,b)$, so by Cor. 29.4, there exists some constant c such that $h(x) = c$ for all $x \in (a,b)$. Thus $f(x) = g(x) + c$ for all $x \in (a,b)$.

$h(x) = c = f(x) - g(x) \Rightarrow f(x) = g(x) + c.$

Increasing & Decreasing

Def. Let f be a real-valued function defined on an interval I. If $x_1, x_2 \in I$ and $x_1 < x_2$ we say

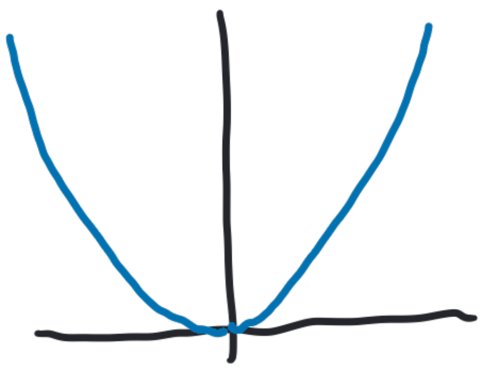
1. f is **increasing** on I if $f(x_1) \leq f(x_2)$ and

   f is decreasing on I if $f(x_1) \geq f(x_2)$

2. f is **strictly increasing** on I if $f(x_1) < f(x_2)$

   f is strictly decreasing on I if $f(x_1) > f(x_2)$

Examples: $f(x) = x^2$



strictly decreasing on $(-\infty, 0]$

strictly increasing on $[0, \infty)$

Another Corollary of the MVT

Corollary 29.7. Let f be a differentiable function on an interval (a, b). Then

1. f is increasing if $f'(x) \geq 0$ for all $x \in (a, b)$.

If $x_1 < x_2$

is $x_1^3 < x_2^3$

2. f is strictly increasing if $f'(x) > 0$ for all $x \in (a, b)$.

T/F - If f is strictly increasing, then $f'(x) > 0$ for all $x \in (a,b)$. - FALSE! $f(x) = x^3$ on $(-1, 1)$

Q: How does this follow from the MVT?